WIS2 Cookbook

World Meteorological Organization

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Chapter 1. Introduction

1.1. Purpose

In conjunction with the Manual on the WMO Information System volume II (WMO-No. 1060) (Manual on WIS volume II: WIS 2.0), the present Guide to the WMO Information System volume II (Guide to WIS volume II: WIS 2.0) is designed to ensure adequate uniformity and standardization in the data, information and communication practices, procedures and specifications employed by Members of the World Meteorological Organization (WMO) in the operation of the WMO Information System WIS 2.0 as it supports the mission of the Organization. The Manual on WIS contains standard and recommended practices, procedures and specifications. The Guide to WIS contains additional information concerning practices, procedures, and specifications that Members are invited to follow or implement in establishing and conducting their arrangements in compliance with the WMO Technical Regulations and in developing meteorological and hydrological services.

This cookbook provides various code snippets, recipes and workflow examples in support of WIS2 requirements.

1.2. Key documents

WIS2 Manuals and Guides can be found at the following locations:

- Manual on the WMO Information System
- Guide to the WMO Information System Volume II - WMO Information System 2.0
- Provisions for the Transition from the WMO Information System (WIS) 1.0 and Global Telecommunication System to WIS 2.0

1.3. Publication and contribution

Updates to this cookbook will be published twice per year - in March and October.

All "recipes" published in this cookbook have been reviewed by experts from the INFCOM Standing Committee on Information Management and Technology (SC-IMT). Publication of a new release is approved by the Chair of SC-IMT.

NOTE	The WIS2 Cookbook is not regulatory material and does not follow the approval and publication process for the WIS2 Manual and Guide.

The WIS2 community are invited to provide feedback on the "recipes" in this Cookbook and also to propose new recipes. Contributions should be made by raising issues in the wmo-im/wis2-cookbook GitHub repository. Experts from SC-IMT will review suggestions with the aim to incorporate them into the next release of the WIS2 Cookbook.

[1] https://wmo.int/site/knowledge-hub/governance/infcom/standing-committee-information-management-and-technology-sc-imt

[2] https://wmo.int/site/knowledge-hub/governance/infcom

Chapter 2. Recipes for data consumers

2.1. Searching the Global Discovery Catalogue

The Global Discovery Catalogue (GDC) allows for a wide range of query predicates to search for data in WIS2 as per the OGC API - Records - Part 1: Core specification.

The GDC can be searched via the `/collections/wis2-discovery-metadata/items` endpoint. This endpoint provides a number query parameters as described in the examples below.

NOTE: examples below are not URL encoded for clarity / readability, but should be when interacting with the GDC.

2.1.1. Spatial queries

- search for metadata records of data in Canada: `bbox=-142,42,-52,84`

Note that the format of `bbox` is comma-separated values in the following order:

- minimum longitude
- minimum latitude
- maximum longitude
- maximum latitude

2.1.2. Temporal queries

- search for metadata records updated since 29 July 2024: `datetime=2024-07-29/..`
- search for metadata records updated before 29 July 2024: `datetime=../2024-07-29`
- search for metadata records updated on 29 July 2024: `datetime=2024-07-29`

2.1.3. Equality queries

- search for metadata records whose title contains the terms hourly observations: `title=hourly observations`
- search for metadata records whose title contains the terms hourly or observations: `title=hourly | observations`
- search for metadata records for a specific contact organization `contacts.organization=Direction Generale de la Météorologie`

2.1.4. Freetext search

- search metadata records for temperature: `q=temperature`
- search metadata records for GRIB2 data: `q=GRIB2`
- search metadata records for any GRIB data: `q=*GRIB*`

- search metadata records for any GRIB data in Germany: `q=*GRIB* AND germany`
- search for either GRIB data or data in Europe with subscriptions to the Météo-France Global Broker: `q=(*GRIB* OR *Europe*) AND *globalbroker*`
- search for data from Belize with MQTT subscription capabilities: `q="cache/a/wis2/bz-nms"`

2.1.5. Sorting

- sort search results by title, ascending: `sortby=title`
- sort search results by title, descending: `sortby=-title`

2.1.6. Paging

- present search results 1-10: `limit=10`
- present search results 11-20: `limit=10&offset=10`
- limit to 3 search results: `limit=3`

2.2. Finding data subscription services from the Global Discovery Catalogue

The Global Discovery Catalogue (GDC) contains both real-time and non real-time data. The WMO Core Metadata Profile (WCMP2) allows for description of real-time data via its distribution information, which data publishers use to describe and define connectivity and subscription information for a given dataset.

A typical WCMP2 distribution link for data subscriptions can be found below:

```
{
    "rel": "items",
    "href": "mqtts://everyone:everyone@globalbroker.meteo.fr:8883",
    "channel": "origin/a/wis2/ca-eccc-msc/data/core/hydrology",
    "type": "application/geo+json",
    "title": "Data notifications"
}
```

NOTE The `channel` property represents WIS2 topic which can be used to subscribe to the `href` property (i.e. the MQTT address) of the Global Broker (GB).

Programmatically, a GDC client can query the catalogue and filter the results for real-time subscriptions in the following manner:

```
import requests

response = requests.get('https://wis2-gdc.weather.gc.ca/collections/wis2-discovery-
metadata/items').json()
```

```
def is_wis2_subscription_link(link) -> bool:
    if (link['href'].startswith('mqtt') and
            link.get('channel', '').startswith(('origin/a/wis2', 'cache/a/wis2'))):
        return True


for feature in response['features']:
    for link in feature['links']:
        if is_wis2_subscription_link(link):
            print('WIS2 subscription link')
```

Using the `href` and `channel` properties of a matching link object, a client can connect and subscribe to data notifications for a given dataset.

Chapter 3. Recipes for data publishers

3.1. WIS2 Node Assessment Process

Data publishers are required to demonstrate that their WIS2 Node implementation meets the requirements of the WIS2 specification as part of the "WIS2 Node Assessment Process".

The WIS2 Node Assessment Process is initiated whenever an authorized data publisher submits a WIS2 Node registration request to the WMO Secretariat.

Global Information System Centres (GISCs) are responsible for assessing WIS2 Node implementations in their Area of Responsibility (AoR) and providing feedback to the WMO Secretariat.

For an overview GISCs and their AoRs, see WMO: WIS centres (NCs, DCPCs, and GISCs) on the WMO Knowledge Hub.

3.1.1. Roles and Responsibilities

Prior to the WIS2 Node Assessment process, the WMO Secretariat will ensure the centre-id is published in the WMO Codes Registry at http://codes.wmo.int/wis/topic-hierarchy/centre-id

During the WIS2 Node assessment:

- The WMO Secretariat connects the WIS2 Node to the WIS2 Development Environment (called wis2dev later).
- The GISC associated to the new WIS2 Node performs the assessment.
- The GISC informs WMO Secretariat when the assessment has been completed.

NOTE

Optional pre-registration testing

Data publishers or WIS2 solution providers who want to test their systems before registration can contact the WMO Secretariat at wis2-support@wmo.int to request that a test system be connected to wis2dev.

3.1.2. WIS2 Development Environment

The WIS2 Development Environment (wis2dev) consists of:

- A Global Broker (wis2dev GB) with a publicly accessible MQTT endpoint available at: `mqtts://gb.wis2dev.io:8883`
- A Global Discovery Catalogue (wis2dev GDC) with an OGC API - Records endpoint available at: `https://gdc.wis2dev.io/`
- A Global Cache (wis2dev GC) provided through Amazon S3 web services

3.1.3. wis2dev GB

When a new WIS2 Node registration request is received from an authorized data publisher, the wis2dev GB is configured with a new subscription to the MQTT endpoint provided by the data publisher.

The new subscription on the wis2dev GB will subscribe to the topic:

- `origin/a/wis2/<centre-id>/#`

where `<centre-id>` is the unique identifier of the WIS2 Node.

The wis2dev GB republishes all valid WIS2 Notification Messages. A message is considered invalid when:

- the topic used when publishing the Notification Message does not match any approved topic in the WIS2 Topic Hierarchy, or
- the message does not conform to the WIS2 Notification Message specification, or
- the message is a data notification for which no corresponding discovery metadata record can be found in the wis2dev GDC.

3.1.4. wis2dev GC

The wis2dev Global Cache downloads data advertised in data notifications published on:

- `origin/a/wis2/<centre-id>/data/core/#`
- `origin/a/wis2/<centre-id>/metadata/#`

No download attempt is made if the data has already been downloaded, unless the download link indicates `rel="update"`.

After a successful download, the wis2dev GC publishes a data notification on `cache/a/wis2/<centre-id>/#`, with the full topic matching that of the original WIS2 Node, but with `origin` replaced by `cache`.

Notifications published by the wis2dev GC on `cache/a/wis2/#` are republished by the wis2dev GB.

3.1.5. wis2dev GDC

The wis2dev Global Discovery Catalogue subscribes to the wis2dev GB at `cache/a/wis2/<centre-id>/metadata` and validates discovery metadata records published by the WIS2 Node using the pywcmp validation and quality assessment tool.

3.1.6. Assessment Preparation for Data Publishers

Before the assessment, data publishers should ensure that:

- The wis2dev IP address, `52.25.203.140`, is allowlisted for the HTTP and MQTT ports used by the WIS2 Node.
- The WIS2 Node can publish data notifications for metadata, and the published metadata

contains a valid WCMP2 record.

- The WIS2 Node can publish data notifications for data, and the data notifications match the published metadata.

3.2. GISC test setup instructions

GISCs subscribe to `mqtts://gb.wis2dev.io:8883` using username/password `everyone/everyone` and set up a connection to the topics:

- `origin/a/wis2/<centre-id>/#`
- `cache/a/wis2/<centre-id>/#`
- `monitor/a/wis2/<centre-id>`

GISCs will coordinate with the contact point for the new WIS2 Node to first publish discovery metadata for their datasets and then to publish data notifications.

Approximately 30 minutes are needed after metadata is published before the Global Broker can validate that published data has corresponding metadata.

GISCs will follow up on errors reported on the monitor topic (`monitor/a/wis2/#`) and assist the contact point of the new WIS2 Node with corrective actions.

3.2.1. Assessment Completion Checklist

The WIS2 Node Assessment is considered complete when:

- One or more metadata records are available for the WIS2 Node when searching `https://gdc.wis2dev.io/collections/wis2-discovery-metadata/items?q=%3Ccentre-id%3E`
- MQTT notifications seen on wis2dev GB at `origin/a/wis2/<centre-id>/metadata`
- MQTT notifications seen on wis2dev GB at `origin/a/wis2/<centre-id>/data/#`
- MQTT notifications seen on wis2dev GB at `cache/a/wis2/<centre-id>/metadata`
- MQTT notifications seen on wis2dev GB at `cache/a/wis2/<centre-id>/data/core/#`

3.2.2. Common Assessment Issues

Issue	Recommended action
Notifications published under `origin/a/wis2/<centre-id>` by the WIS2 Node are not being republished by the wis2dev GB.	The wis2dev GB will only republish notifications adhering to WIS2 standards. Messages reusing a previously used 'id' will be discarded as duplicates. Ensure that you use a valid WIS2 topic for the MQTT channel, that the payload adheres to the WIS2 Notification Message standard, and that a unique id is used for each notification.

Issue	Recommended action
Canonical URL embedded in a WIS2 data-notification is not accessible from wis2dev	Ensure that the canonical URL is reachable from the wis2dev environment over the public Internet. Check that the DNS resolves to the correct IP address and that the wis2dev incoming IP address, `52.25.203.140`, is allowed to access the WIS2 Node for MQTT(S) / WS(S) and HTTP(S).
WIS2 Node is not publishing data notifications for metadata	Ensure that data notifications for metadata are being published every 24 hours on the topic `origin/a/wis2/<centre-id>/metadata` and that the notification contains a valid WCMP2 record in the data payload.
Publishing a metadata-notification results in a notification on `origin/a/wis2/<centre-id>/metadata` but does not result in a notification on `cache/a/wis2/<centre-id>/metadata` on the wis2dev GB	Updated metadata records are rejected by the wis2dev GC if a previous record with the same data-id was already cached and the download link did not indicate rel="update". Ensure that the download link for updated metadata records indicates rel="update" or the Notification Message is using a new value for `properties.data_id`.
Metadata records published by the WIS2 Node are cached by the wis2dev GC, but rejected by the wis2dev GDC	The wis2dev GDC will reject data that fails the pywcmp validation and quality assessment tool. Non-wis2box users should ensure that the metadata record passes the pywcmp ETS validation prior to publishing (see https://github.com/World-Meteorological-Organization/pywcmp). Data publishers using wis2box should ensure they use the latest version of the wis2box software to ensure the latest version of pywcmp is used. Note that the wis2dev GDC provides pywcmp validation and quality assessment as an online service (see Validating a WMO Core Metadata Profile record)

NOTE

Implementation note regarding wis2box and metadata management

`wis2box` reduces metadata-related issues by enforcing a "dataset-driven workflow".

- Before publishing any data notifications, a dataset must first be created
- To create a dataset, a minimum amount of information must be provided to create the WCMP2 record
- Before publishing the WCMP2 record, the content is always validated using pywcmp
- Data notifications are associated to metadata programmatically by querying a local copy of the published WCMP2 records

WIS2 Node implementations that use separate software for metadata and data notifications are more likely to encounter metadata related issues.

To reduce these issues, follow a wis2box-like workflow: create the metadata record

> first, then publish data notifications based on the published metadata records for your WIS2 Node.

3.3. Best practices for publishing station based observational data

The WMO Unified Data Policy defines the following data types:

- Core Weather Surface-based data: observations provided by the Global Basic Observing Network (GBON) and other observational data, as specified in the Manual on the WMO Integrated Global Observing System (WMO-No. 1160);
- Recommended Weather Surface-based data: all available observations provided by the Regional Basic Observing Network (RBON), which is further specified in the Manual on the WMO Integrated Global Observing System (WMO-No. 1160);
- Core Climate observational data
  - Measurements provided by the Global Climate Observing System (GCOS) Upper-Air Network (GUAN) and GCOS Surface Network (GSN) stations (see also 1.1.1 (a));
  - Climate data as defined in the Manual on High-quality Global Data Management Framework for Climate (WMO-No. 1238);
  - Essential Climate Variables (ECVs) as defined by the GCOS in the Manual on the WMO Integrated Global Observing System (WMO-No. 1160) to the extent that the Member holds the data in a digital archive

Typically, observational data is published using BUFR as the representation/format, as specified in the Manual on Codes (WMO-No. 306). The Manual on the Global Telecommunication System (WMO-No 386) mandates the provision of the data as a "bulletin" (multiple observational data grouped into a single BUFR file).

It is considered that with modern telecommunication protocols, the "bulletin" approach is no longer required. Therefore, in the WIS 2.0 context, the "bulletin" approach is deprecated and observational data should be provided as individual BUFR files without grouping.

As publishing data to WIS 2.0 requires the WIS 2.0 Notificiation Message (WNM), the following aspects should be included when publishing observational data.

3.3.1. Geometry

WNM requires a `geometry` property, which allows for expressing the location of a data granule (in this case a BUFR file) as follows:

```
{
  "geometry": {
    "type": "Point",
    "coordinates": [-87.42188, 44.83941, 712]
  }
```

```
}
```

Note that the order of the geometry coordinates are longitude, latitude and elevation. In addition, when a geometry cannot be provided, `geometry` can be set to `null`.

3.3.2. Client Side Filtering

WNM allows for extensibility (i.e. adding additional properties) by design. To enable client side filtering, and when the WIGOS Station Identifier (WSI) exists for observational data, a WNM should include `properties.wigos_station_identifier` as follows:

```
{
  "properties": {
    "wigos_station_identifier": "0-20000-0-71628"
  }
}
```

By providing geometry and WIGOS station identifier in a WNM for observational data, this allows for users to assess and filter WIS 2.0 notifications using location and WSI as the filter criteria.

For more information on client side filtering see recipe Advertising client side filters for data subscriptions in WCMP2 and WNM.

3.4. Publishing aeronautical meteorology data on WIS2

In addition to publishing aeronautical meteorology data via ICAO SWIM (as defined in the ICAO Procedures for Air Navigation Services - Information Management, Doc 10199), WIS2 may be used as a secondary mechanism to share this data.

However, as per the current version of ICAO Annex 3 - Meteorological Service for International Air Navigation, aeronautical meteorology data may only be used for the purposes of international air navigation.

If you choose to publish aeronautical meteorology data through WIS2 you must clearly state the restrictions on use of this data in the associated WCMP2 discovery metadata record.

First, because aeronautical meteorology data is subject to usage restrictions the Recommended data policy applies, as described in the WMO Unified Data Policy, Resolution 1 (Cg-Ext(2021)).

Second, the data usage rights must be stated.

Third, because we're using the Recommended data policy, you must provide a license for use of the dataset. This license must be published so that it can be referenced from a Link.

Proposed clauses in license for use of aeronautical meteorological data:

```
AERONAUTICAL METEOROLOGY DATA LICENSE
```

```
  TERMS OF USE

  This aeronautical meteorological data is provided under the following terms:

    PERMITTED USE

    This meteorological information is made available exclusively for the purposes of
safety of air navigation.

    RESTRICTIONS

    Air Navigation Use Only: This data shall only be used for aviation-related
purposes.

    COMPLIANCE

    Use of this data constitutes acceptance of these terms.
```

Describing an aeronautical meteorology dataset:

```
{
    ...
    "properties":{
        "wmo:dataPolicy": "recommended",
        "rights": "This information is freely disseminated for the purposes of safety
of air navigation. ICAO Annex 3",
        ...
    },
    "links": [{
        "rel": "license",
        "href": "https://example.org/license",
        "type": "text/html",
        "title": "Aeronautical Meteorological Data License"
    }],
    ...
}
```

WIS2 Notification Messages indicating the availability of aeronautical meteorology data should be published to the `weather/aviation` topic or sub-topic thereof as stated in the WMO Codes Registry [1].

Example WIS2 Topic for publishing aeronautical meteorology data:

```
origin/a/wis2/{centre-id}/data/recommended/weather/aviation/metar
```

You may restrict access to the aeronautical meteorology data you publish through WIS2, for example, requiring authenticated access. Please refer to recipe Publishing a WIS2 Notification Message with access control for further details.

Please be aware that if you choose to embed aeronautical meteorology data within WIS2

Notification Messages, this data will be openly visible within WIS2. For more information about embedding data in notification messages, please refer to recipe Publishing a WIS2 Notification Message with embedded data.

3.5. Publishing a WMO Core Metadata Profile record for historical data

WIS2 and WCMP2 have been designed to support both real-time and historical data. While real-time data publishing in WIS2 typically includes real-time data notifications, archive data may or may not require data notifications, depending on how often the dataset is disseminated and/or updated.

3.5.1. Temporal aspects

Of critical importance is describing the temporal nature of the archive. Some examples:

Describing the temporal nature of an archive in WCMP2 which contains various temporal granularities

```
"time": {
    "interval": [
        "1850-01-01",
        ".."
    ]
}
```

Describing the temporal nature of an archive in WCMP2 which has a single temporal granularity (i.e. continuously updated on a daily basis)

```
"time": {
    "interval": [
        "1850-01-01",
        ".."
    ],
    "resolution": "P1D"
}
```

Describing the temporal nature of an archive in WCMP2 which is not updated

```
"time": {
    "interval": [
        "1850-01-01",
        "2025-08-19"
    ],
    "resolution": "P1D"
}
```

3.5.2. Distribution and notification aspects

The way in which a historical archive is disseminated can result in a few different possibilities in WCMP2 and WIS2 data notifications.

3.5.2.1. Publishing links of an archive in WCMP2 provided via an API

Historical data made available via an Application Programming Interface (API) allows for data subsetting and reduction (i.e. extracting archive data based on space, time or other attributes), for users to focus on and work with only the part of the archive they are interested in.

While WIS2 does not currently mandate specific APIs, useful APIs for data access include (but are not limited to):

- OGC API - Environmental Data Retrieval: lightweight interfaces to query and access environmental data resources
- SpatioTemporal Asset Catalog API: an API to retrieve data granules

Note: APIs may implement access control as required (see the Access control section in WCMP2).

3.5.2.2. Publishing links of an archive in WCMP2 whose updates are not disseminated as data granules

WIS2 Notifications promote "actionable links" when describing data updates via MQTT. In some cases, a historical archive may be updated without publishing data granule level updates. For example, if an archive is updated every 90 days, and updates are published into the existing archive without the ability to advertise one WIS2 Notification per data granule update, then publishing via MQTT is not a suitable mechanism for real-time decision support systems and downstream applications. In this case, clearly describing the temporal nature of the archive as well as a link which clearly defines an archive (via the `rel=archives` link relation) is recommended:

Publishing an archive link in WCMP2

```
{
    "rel": "archives",
    "href": "https://example.org/hydrometric-archive",
    "type": "text/html",
    "title": "Hydrometric archive database download",
    "distribution": {
        "availableFormats": [
            {
                "name": "SQLite3"
            },
            {
                "name": "Microsoft Access"
            }
        ]
    }
}
```

3.5.2.3. Publishing links via updates into an existing archive

If a historical archives does publish data granules (i.e. if a daily climate archive is updated daily and each day's update can be defined with an actionable link, then an MQTT link is also recommended.

Publishing archive and MQTT links in WCMP2 with data granule updates

```
{
    "rel": "archives",
    "href": "https://example.org/daily-climate-observations",
    "type": "text/csv",
    "title": "Raw data download (CSV files)"
},
{
    "rel": "items",
    "href": "mqtts://example.org",
    "channel": "origin/a/wis2/ca-eccc-msc/data/core/climate/surface-based-
observations/daily",
    "type": "application/geo+json",
    "title": "Data notifications of daily updates"
}
```

3.5.2.4. Publishing notifications of discovery metadata (WCMP2) considerations

Data on WIS2 requires discovery metadata (WCMP2). For data publishers of historical archive data without the required WIS2 Node broker (MQTT) capability, there exist some options to lower the barrier to publication:

- No-code WCMP2 generation / publishing: using GitHub, a data publisher can manage metadata and trigger WCMP2 generation and publication via MQTT using GitHub Actions. This GitHub repository provides a GitOps style workflow for easy management of metadata records as YAML configurations using pygeometa "Metadata Control Files (MCF)". As MCF files are updated, GitHub Actions are triggered to generate WCMP2 records and publish them to a pre-defined MQTT broker that is connected to WIS2. Note that MCF files are simply a compact and abstract approach to manage metadata. A data publisher can also manage WCMP2 directly and publish to a pre-defined MQTT broker in the same manner (as demonstrated by the ECCC/MSC WIS2 Node implementation).
- GISC collaboration: a GISC can support its Area of Responsibility (AoR) by managing the entire metadata publication process, or enable metadata publishing to its MQTT broker to help Members.

Note that the above options are not exhaustive, and additional approaches may be implemented to meet data/metadata publishing requirements of WIS2.

3.6. Publishing a WMO Core Metadata Profile record for retired data

The primary target for WCMP2 records is for real-time or archive data discovery, access and

visualization, with the understanding that a dataset (and its associated discovery metadata) is properly managed and kept up to date. In certain cases, datasets may become unavailable; this could mean that the dataset lifecycle has reached end of life, and/or no longer maintained.

An "end of life" or retired dataset should be independent from its description. That is, even if a dataset is retired, there is value in having discovery metadata for these datasets, and so the WCMP2 record associated with it should not be deleted from the WIS2 Global Discovery Catalogues. The FAIR Data Principles (A2) also state that metradata should persist even the data are no longer sustained [2].

To describe a retired dataset in WCMP2, one can use the `properties.status` object as per below:

Describing a retired dataset in WCMP2

```
"properties": {
  "status": {
    "id": "retired",
    "title": "Retired",
    "url": "https://standards.iso.org/iso/19115/-
3/mcc/1.0/codelists.xml#MD_ProgressCode_retired"
  }
}
```

The above status is defined by ISO 19115-3 as "*item is no longer recommended for use. It has not been superseded by another item*" [3].

Another aspect to consider reviewing all links in the WCMP2 record (`links` object). For example, MQTT links can be removed, and any links which no longer provide the data can also be removed. If the data is retired and has alternative access mechanisms (email, web form), these links can be added to the record.

3.7. Publishing a WMO Core Metadata Profile record with data provenance

Data provenance provides the ability to describe an audit trail for the lifecycle of a given dataset.

In WCMP2, one can provide linkages to provenance information via a dedicated link object.

Expressing data provenance as a WCMP2 link

```
{
    "rel": "describedby",
    "type": "text/html",
    "title": "Provenance information for this dataset",
    "href": "https://example.org/dataset1.html"
}
```

As with any WCMP2 link, the URL itself is defined by the data publisher. The `describedby` attribute

(which is an official IANA link relation) is the signal for identifying data provenance by a user assessing and reviewing dataset history and lineage.

3.8. Publishing a WMO Core Metadata Profile record referencing WIGOS stations and networks

WCMP2 allows for publication of any data (observations, forecasts, warnings, etc.) with the principle of discovery metadata. WIGOS metadata refers to station or observation metadata which is currently managed in the OSCAR Surface system.

Given that various metadata are managed across distributed systems, WCMP2 links can be used to relate a given dataset of observations measured at a station or set of stations on OSCAR/Surface, or other/similar systems.

Referring to a station as a WCMP2 link

```
{
    "rel": "station",
    "type": "application/xml",
    "title": "Station report details for Thessaloniki (Greece)",
    "href": "https://oscar.wmo.int/surface/rest/api/wmd/download/0-20008-0-THE"
}
```

Referring to a set of stations as a WCMP2 link

```
{
    "rel": "stations",
    "type": "text/html",
    "title": "Davos Integrated CryoNet Cluster",
    "href": "https://oscar.wmo.int/surface/#/search/stationClusterReport/8"
}
```

A data publisher can decide whether to enumerate one link foreach station, or refer to a facility set of stations. As an example, for a network of 50 stations, it would be more suitable to refer to the set of stations as a single `rel=stations` link.

NOTE	Link relations are governed by IANA link relations and WMO extensions as defined in WCMP2, available on the WMO Codes Registry.

3.9. Publishing a WIS2 Notification Message with access control

Recommended data in WIS2 may be open or access controlled. For data publication with access control implications, WNM provides a `security` object as part a link object. The `security` object is defined using OpenAPI Security Scheme definitions.

Access control using HTTP Basic authentication

```
{
  "rel": "canonical",
  "type": "application/grib2",
  "href": "https://example.org/my/protected/data/nwp/12/003/20240805120000-air-temp-
500.grib2",
  "security": {
    "default": {
      "type": "http",
      "scheme": "basic",
      "description": "Please contact us for access information"
    }
  }
}
```

Access control using an API key

```
{
  "rel": "canonical",
  "type": "application/geo+json",
  "href": "https://example.org/my/protected/data/nwp/12/003/20240805120000-air-temp-
500.grib2",
  "security": {
    "default": {
      "type": "apiKey",
      "name": "api-key",
      "in": "query",
      "description": "Please see https://example.org/contact-us for more information"
    }
  }
}
```

Note:

- the child property under `security` (`default` in the examples above) can be any text or label. We use `default` here as a convention
- the `security.default.name` is the name of the API key parameter as defined by your API service
- only properties defined in the OpenAPI Security Scheme definition are allowed. Any additional properties will invalidate the WNM

Of course, always ensure your WNM is valid (see [Validate a WIS2 Notification Message] for more information).

3.10. Publishing a WIS2 Notification Message with embedded data

A WIS2 Notification Message provides the ability to embed data as part of a notification (see the Properties / Content section of the specification).

Providing embedded data offers users the ability to access the data without the need to execute the additional step of downloading the data via the actionable link (see the Links section of the specification).

In some cases, a data publisher also prefer not to publish the corresponding actionable link to infrastructure (e.g. an HTTP or FTP server).

When the data publisher decides to not publish an actionable link and to rely exclusively on the embedded data feature, the data publisher will:

- use `properties.content` object to provide the data
- use `properties.cache=false` to instruct WIS2 Global Cache Services to not cache the data (note: `properties.cache` is not required for recommended data, since recommended data is not cached by WIS2 Global Caches)
- provide a link object whose `href` results in HTTP 204 (No Content)

To provide an HTTP 204 link, there a couple of options:

Option 1: configure your web server to provide an endpoint that always returns 204:

Nginx example configuration

```
# always return HTTP 204 when accessing /no-content-here
location /no-content-here/ {
  return 204;
}
```

Apache example configuration

```
# always return HTTP 204 when accessing /no-content-here
Redirect 204 no-content-here
```

which then would result in the link object as follows:

```
{
    "href": "https://example.org/no-content-here",
    "rel": "canonical"
}
```

Option 2: use a service that provides HTTP responses as a service. One such service is the HTTP Response API, which then would result in the link object as follows:

```
{
    "href": "https://http.codes/204",
    "rel": "canonical"
}
```

Embedded data example:

```
{
    "id": "31e9d66a-cd83-4174-9429-b932f1abe1be",
    "conformsTo": [
        "http://wis.wmo.int/spec/wnm/1/conf/core"
    ],
    "type": "Feature",
    "geometry": {
        "type": "Point",
        "coordinates": [
            6.146255135536194,
            46.223296618227444
        ]
    },
    "properties": {
        "pubtime": "2022-03-20T04:50:18Z",
        "datetime": "2022-03-20T04:45:00Z",
        "integrity": {
            "method": "sha512",
            "value": "A2KNxvks...S8qfSCw=="
        },
        "cache": false,
        "data_id": "dataset/123/data-granule/UANT01_CWAO_200445___15103.bufr4",
        "metadata_id": "urn:wmo:md:ca-eccc-msc:observations.swob",
        "content": {
            "encoding": "utf-8",
            "value": "encoded bytes from the file",
            "size": 457
        }
    },
    "links": [
        {
            "href": "https://http.codes/204",
            "rel": "canonical"
        }
    ]
}
```

For cases of publishing recommended data, the same approaches/example can be used, without the need to set `properties.cache`.

3.11. Publishing a WIS2 Notification Message for resource deletion

A WIS2 Notification Message provides the ability to publish notifications for new, updated or deleted data and metadata (see the Links section of the specification).

Similar to the embedded data Recipe, for data or metadata deletions (core or recommended data), a data publisher may not wish to manage or publish the data or metadata link to infrastructure to an HTTP or FTP server.

In this case, a similar strategy can be used to provide an HTTP 204 No Content link, as per the embedded data Recipe, along with setting `rel=deletion` in the link object.

Resource deletion example:

```
{
    "id": "31e9d66a-cd83-4174-9429-b932f1abe1be",
    "conformsTo": [
        "http://wis.wmo.int/spec/wnm/1/conf/core"
    ],
    "type": "Feature",
    "geometry": {
        "type": "Point",
        "coordinates": [
            6.146255135536194,
            46.223296618227444
        ]
    },
    "properties": {
        "pubtime": "2022-03-20T04:50:18Z",
        "datetime": "2022-03-20T04:45:00Z",
        "integrity": {
            "method": "sha512",
            "value": "A2KNxvks...S8qfSCw=="
        },
        "cache": false,
        "data_id": "dataset/123/data-granule/UANT01_CWAO_200445___15103.bufr4",
        "metadata_id": "urn:wmo:md:ca-eccc-msc:observations.swob"
    },
    "links": [
        {
            "href": "https://http.codes/204",
            "rel": "deletion"
        }
    ]
}
```

3.12. Advertising client side filters for data subscriptions in WCMP2 and WNM

A key concept of a WCMP2 record is "actionable links"; this means being able to access a dataset or data granule without any further interactions. For real-time data, a WCMP2 record provides linkages to the WIS2 Global Broker via the MQTT protocol. At its core, MQTT has two key components:

- topic: the topic to subscribe to
- message payload: the message provided as part of a notification to a given topic

WIS2 defines the WIS2 Topic Hierarchy (WTH) and WIS2 Notification Message (WNM) standards which provide a standards-based GeoJSON payload/message.

A typical MQTT link in a WCMP2 document is defined as follows:

Typical WCMP2 MQTT link

```
{
  "rel"  : "items",
  "type" : "application/geo+json",
  "title": "WIS2 notification service",
  "href" : "mqtts://example.org",
  "channel": "cache/a/wis2/ca-eccc-msc/data/core/weather/surface-based-
observations/synop"
}
```

Given WCMP2, WTH and WNM, a user can subscribe to topics related to data of interest for download and access.

In some cases, a dataset may be organized in a manner which requires additional further "filtering" such that a data consumer is only interested in a certain subset of the data granules being advertised by a given WNM. Some examples include (but are not limited to), where a data consumer may be only be interested in:

- surface weather observations from a certain station, or
- numerical weather prediction forecast data for a certain timestep or weather parameter

To implement this behaviour, add additional properties to both WCMP2 and WNM as follows:

3.12.1. Example: Surface weather observations

Surface weather observations: WCMP2 MQTT link with additional properties

```
{
  "rel"  : "items",
  "type" : "application/geo+json",
  "title": "Real-time notifications",
```

```
  "href" : "mqtts://globalbroker.meteo.fr:8883",
  "channel": "cache/a/wis2/ca-eccc-msc/data/core/weather/surface-based-
observations/synop",
  "filters": {
    "wigos_station_identifier": {
       "type": "string",
       "title": "WIGOS station identifier"
    }
  }
}
```

Surface weather observations: WNM additional properties

```
{
  "properties": {
    "wigos_station_identifier": "0-20000-0-71628"
  }
}
```

When implemented by a data producer, a data consumer can:

- subscribe to real-time notifications to the given topic
- perform client side filtering by against all incoming WNMs with `properties.wigos_station_identifier = "0-20000-0-71628"`

3.12.2. Example: Numerical weather prediction based forecast

Numerical weather prediction: WCMP2 MQTT link with additional properties

```
{
  "rel"  : "items",
  "type" : "application/geo+json",
  "title": "Real-time notifications",
  "href" : "mqtts://globalbroker.meteo.fr:8883",
  "channel": "origin/a/wis2/ca-eccc-msc/data/core/weather/prediction/forecast/medium-
range/deterministic/global",
  "filters": {
    "model_run": {
       "type": "string",
       "title": "Model run",
       "enum": [
           "00",
           "12"
        ],
        "example": "00"
    },
    "forecast_hour": {
       "type": "string",
       "title": "Forecast hour",
```

```
        "example": "004"
    }
  }
}
```

Numerical weather prediction: WNM additional properties

```
{
  "properties": {
    "model_run": "00",
    "forecast_hour": "004"
  }
}
```

A data producer would extend WCMP2 and WNM as follows:

- WCMP2: add a link `properties` object for MQTT links, where each key of the link `properties` object is a JSON Schema property definition
- WNM: add additional properties (key: value pairs) in the `properties` object as desired

When implemented by a data producer, a data consumer can:

- subscribe to real-time notifications to the given topic
- perform client side filtering against all incoming WNMs with `properties.model_run = "00"` and `properties.forecast_hour = "004"`

A sample Python script can be found below. The script connects to the Météo-France Global Broker, subscribed to weather notifications from Environment and Climate Change Canada, Meteorological Service of Canada. The script then performs client side filtering by evaluating (for each WNM) the `properties.wigos_station_identifier` value to match a particular station (`0-20000-0-71628`).

Sample Python script to perform client side filtering

```
import json
from paho.mqtt import client as mqtt_client

broker = 'globalbroker.meteo.fr'
port = 8883
username = 'everyone'
password = 'everyone'
topic = 'cache/a/wis2/ca-eccc-msc/data/core/weather/surface-based-observations/synop'

wsi_to_filter = '0-20000-0-71628'


def connect_mqtt() -> mqtt_client:
    def on_connect(client, userdata, flags, reason_code, properties):
        if reason_code == 0:
            print(f'Connected to {broker}')
```

```
        else:
            print(f'Failed to connect: {reason_code}')

    def on_log(client, userdata, level, message):
        print("LOG:", message)

    client = mqtt_client.Client(mqtt_client.CallbackAPIVersion.VERSION2,
                                client_id='s123')
    client.username_pw_set(username, password)
    client.on_connect = on_connect
    client.on_log = on_log
    client.tls_set(tls_version=2)
    client.connect(broker, port)

    return client


def subscribe(client: mqtt_client):
    def on_message(client, userdata, message):
        message_dict = json.loads(message.payload.decode())

        print('Performing client side filtering')
        wsi = message_dict['properties'].get('wigos_station_identifier')

        if wsi != wsi_to_filter:
            print(f'Topic: {message.topic}')
            print(f'Payload: {message.payload.decode()}')

    client.subscribe(topic)
    client.on_message = on_message


def run():
    client = connect_mqtt()
    subscribe(client)
    client.loop_forever()


if __name__ == '__main__':
    run()
```

3.13. Validating a WMO Core Metadata Profile record

The pywcmp tool provides a test suite to validate a message against the WCMP2 specification requirements, as well as a Python API for application integration. Consult the pywcmp README on GitHub for more information/examples.

Using pywcmp for WCMP2 validation

```
# install pywcmp
```

```
pip3 install pywcmp

# sync WCMP2 schemas and codelists
pywcmp bundle sync

# validate WCMP2 against abstract test suite (file on disk)
pywcmp ets validate /path/to/file.json

# validate WCMP2 against abstract test suite (URL)
pywcmp ets validate https://example.org/path/to/file.json
```

A WCMP2 record can also be validated using pywcmp "as a service" using the Canadian and German WIS2 Global Discovery Catalogues, which provide an online validator:

- Navigate to https://wis2-gdc.weather.gc.ca/openapi?f=html or https://wis2.dwd.de/gdc/openapi?f=html
- Navigate to section **pywcmp-wis2-wcmp2-ets**, endpoint `/processes/pywcmp-wis2-wcmp2-ets/execution` (POST)
- Click "Try it out"
- In the section "Mandatory execute request JSON", paste the WCMP2 JSON inside the `record` object

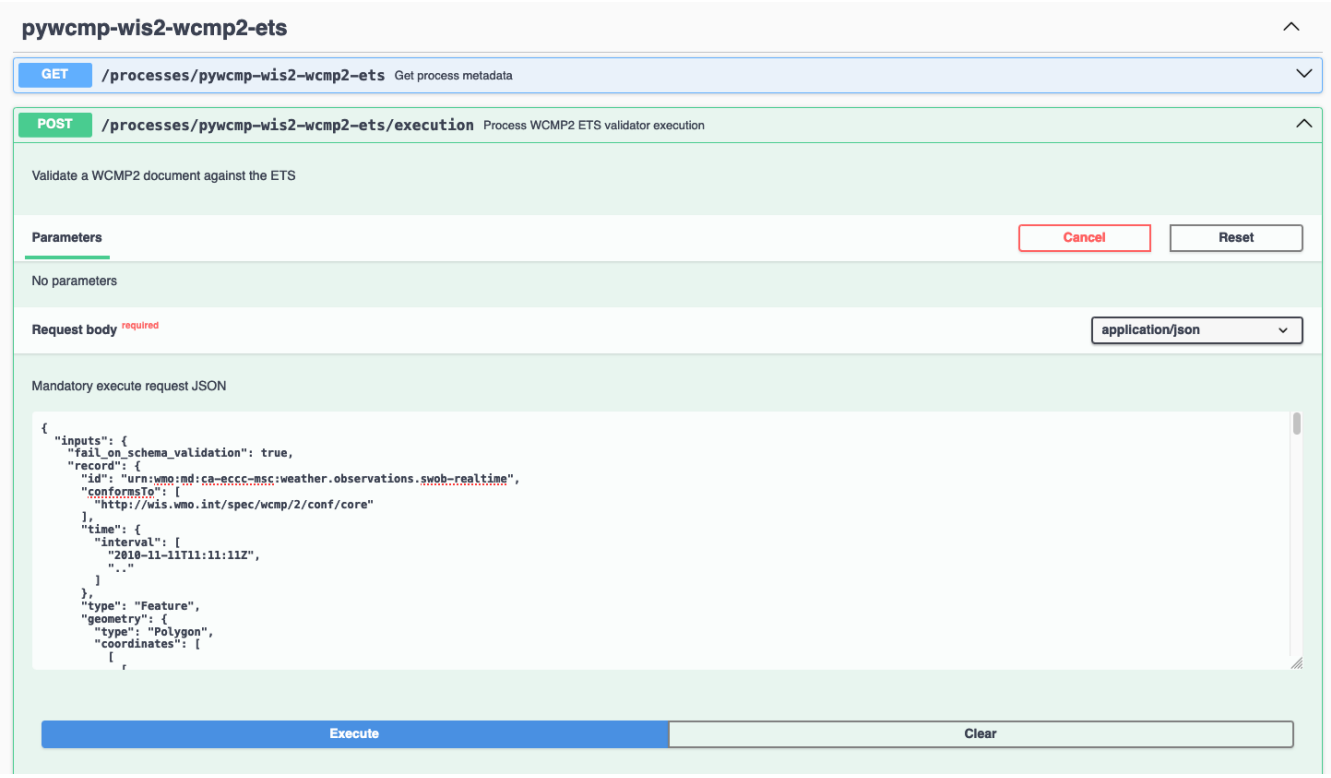


Figure 1. WIS2 GDC online validator, request via WCMP2 copy/paste

Alternatively, the validator service also accepts a URL (if the WCMP2 record is online):

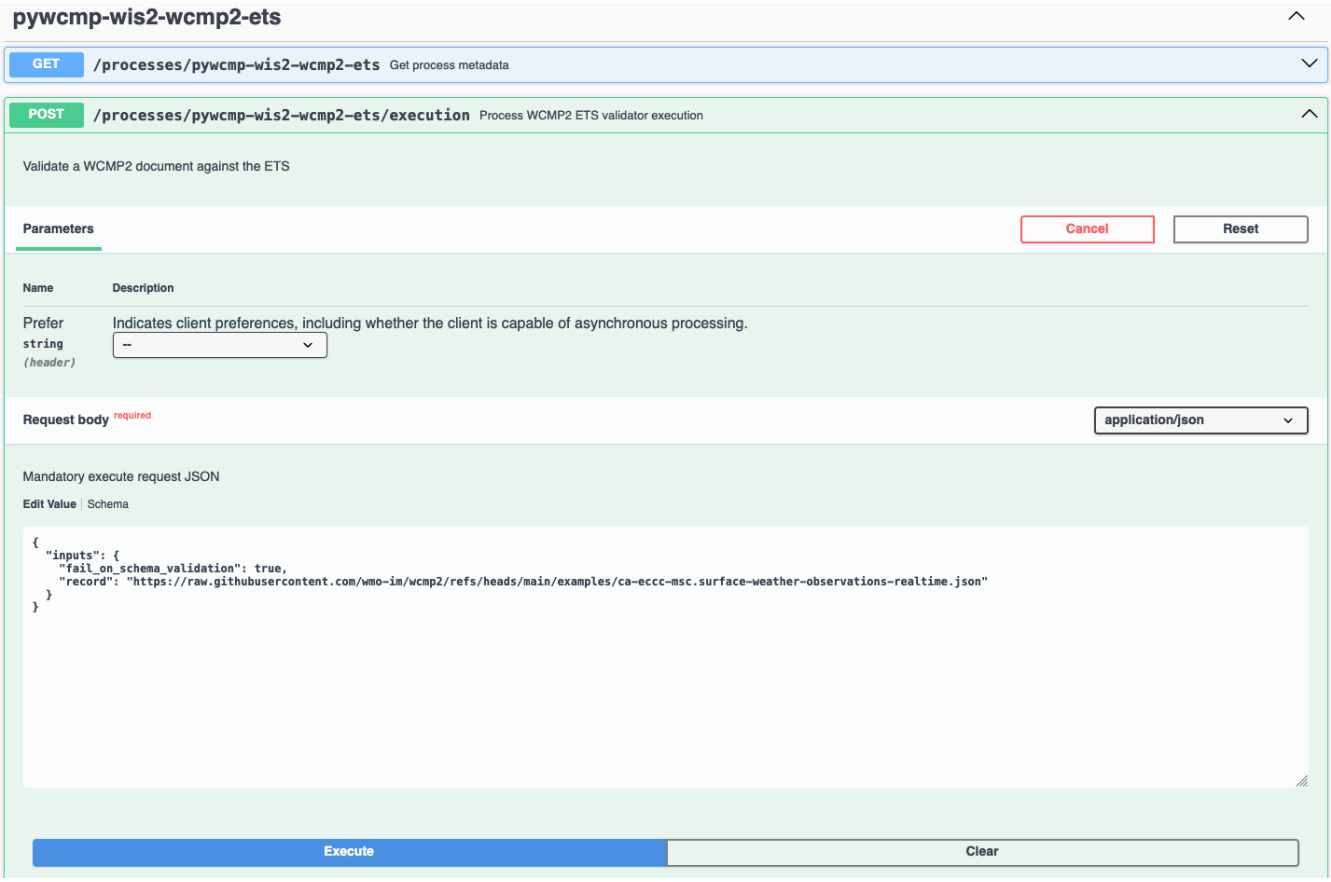


Figure 2. WIS2 GDC online validator, request via WCMP2 URL

- Click "Execute"

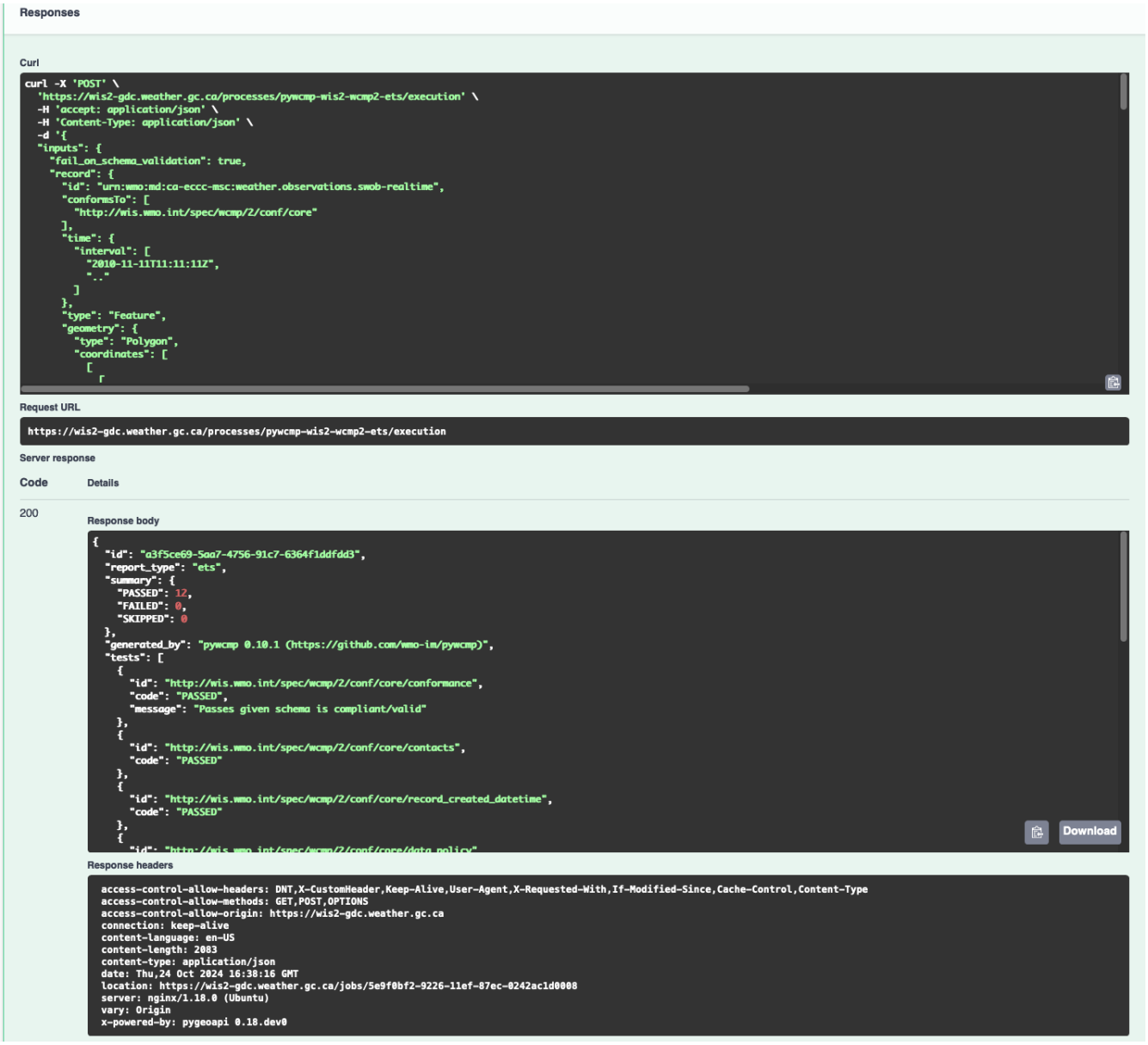


Figure 3. WIS2 GDC online validator, response

A response will be provided with validation results.

The WCMP2 standard also has (draft) Key Performance Indicators that provide quality assessment in support of continuous improvement of WCMP2 records. pywcmp additionally implements these KPIs using a similar workflow.

Using pywcmp for WCMP2 quality assessment

```
# validate WCMP2 against abstract test suite (file on disk)
pywcmp kpi validate /path/to/file.json

# validate WCMP2 against abstract test suite (URL)
pywcmp kpi validate https://example.org/path/to/file.json
```

The KPI check performs actions such as checking for healthy/working links and acronym checks, length of titles and more. Note that the KPI results do not affect WCMP2 compliance but are suggestions on how to improve your WCMP2 record for better discoverability and use.

The Canadian and German WIS2 Global Discovery Catalogues also provide this functionality online in a similar fashion as the validator.:

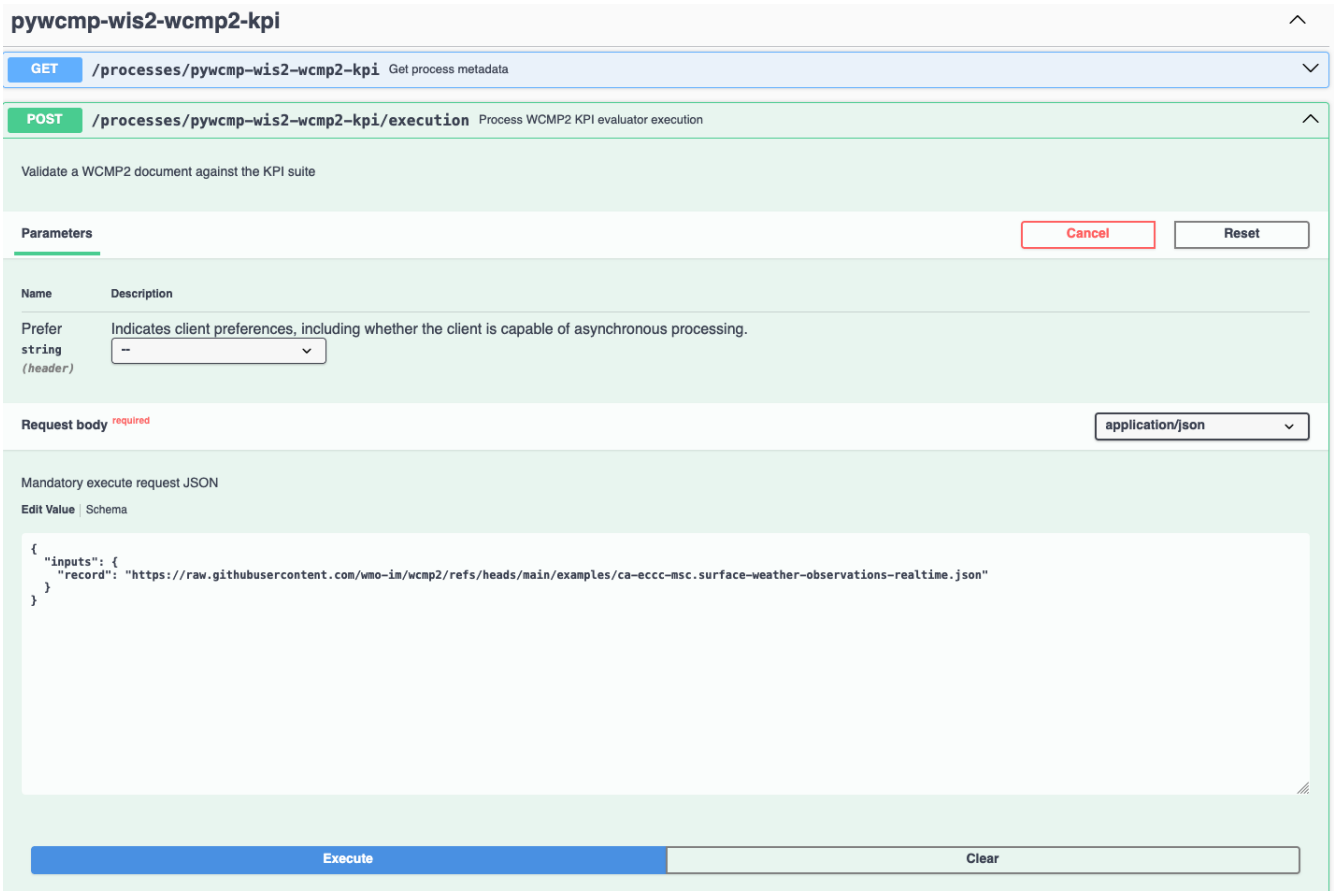


Figure 4. WIS2 GDC quality assessment checker, request via WCMP2 URL

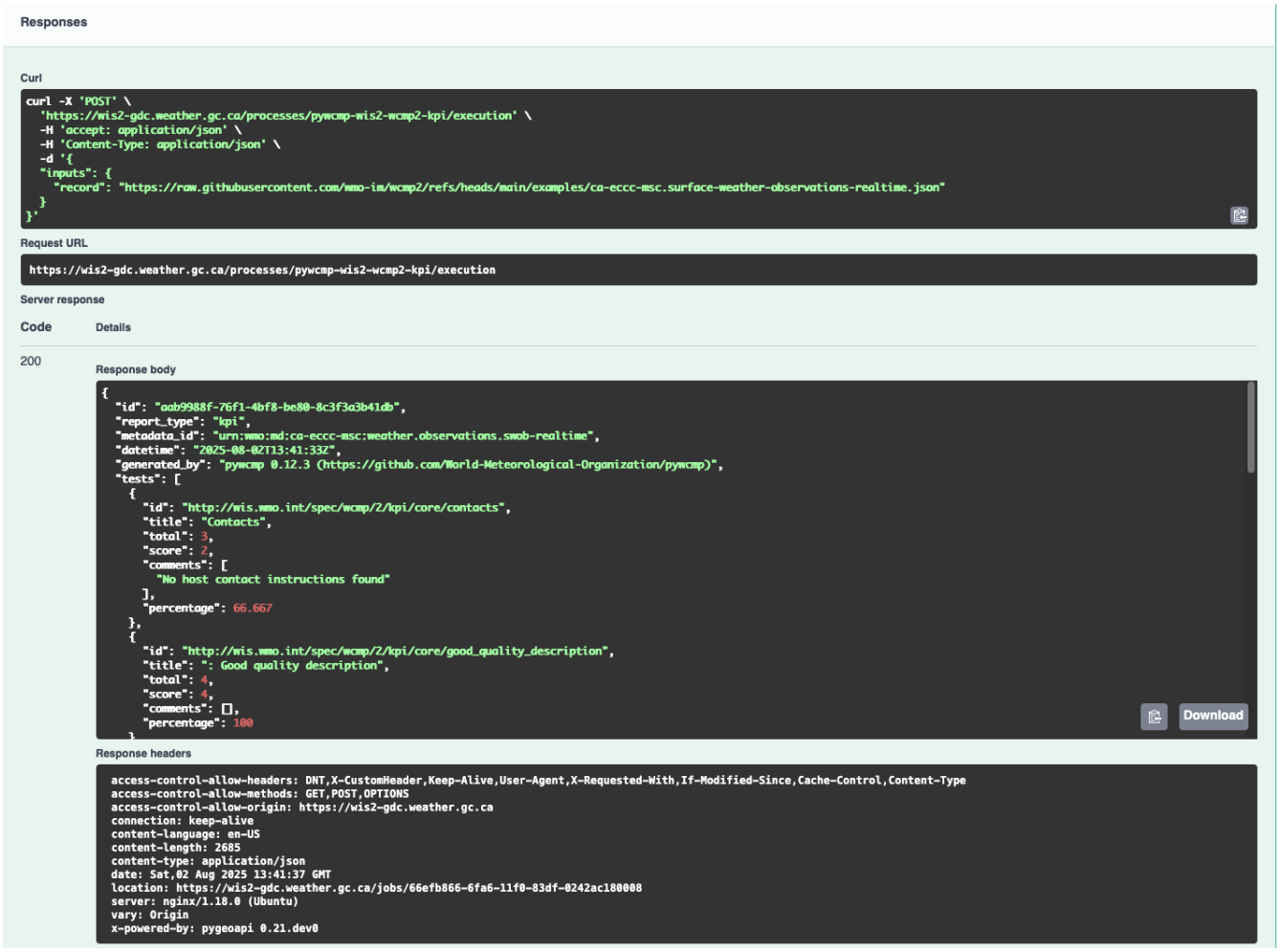


Figure 5. WIS2 GDC quality assessment checker, response

A response will be provided with quality assessment results (in a similar format to the validation report).

3.14. Validating a WIS2 Notification Message

A WIS2 Notification Message provides a JSON Schema which can be used by any programming language that supports JSON and JSON Schema validation.

Using Python and check-jsonschema

```
# install check-jsonschema Python Package from the Python Package Index (PyPI)
pip3 install check-jsonschema

# download WNM schema
curl -O http://schemas.wmo.int/wnm/1.1.0/schemas/wis2-notification-message-
bundled.json

# run schema validation
check-jsonschema --schemafile wis2-notification-message-bundled.json
/path/to/my/wnm.json
```

The pywis-pubsub tool provides a test suite to validate a message against the WNM specification

requirements, as well as a Python API for application integration. Consult the pywis-pubsub README on GitHub for more information/examples.

Using pywis-pubsub

```
# install pywis-pubsub
pip3 install pywis-pubsub

# sync WIS2 notification schema
pywis-pubsub schema sync

# validate WNM against abstract test suite (file on disk)
pywis-pubsub ets validate /path/to/file.json

# validate WNM against abstract test suite (URL)
pywis-pubsub ets validate https://example.org/path/to/file.json
```

3.15. Subscribing to GDC reports of WCMP2 validation and quality assessment

When a WIS2 Node publishes a WCMP2 record, GDCs perform validation according to the WCMP2 Abstract Test Suite). If the record is compliant, a GDC will ingest and publish the record. In addition, a GDC may provide additional quality assessment testing based on the (draft) WCMP2 Key Performance Indicators (KPIs).

GDC reports are made available by some GDCs in pilot mode as part of the (draft) WIS2 Monitoring Events specification (WME) and can be found by subscribing to a Global Broker using the topic `monitor/a/wis2/CENTRE_ID`, where:

- `CENTRE_ID` is the centre identifier of the subject of the message

For example, to subscribe to GDC reports on WCMP2 records published by Japan Meteorological Agency (JMA), one would subscribe to the topic `monitor/a/wis2/jp-jma`.

Message payloads are based on WME Message Encoding (WMEM), with a `data` payload of the WCMP2 ETS/KPI reports.

3.16. Providing a requirements specification for a WIS 2.0 Node

The below recipe provides user requirements specifications for the implementation of a WIS 2.0 Node.

3.16.1. WIS 2.0 Node - User requirements specifications

3.16.1.1. Introduction

Applicable Documents

	Document Title	Reference
AD-0	Manual on the WMO Information System, Volume II - WMO Information System 2.0	https://library.wmo.int/idurl/4/68731

Reference Documents

	Document Title	Reference
RD-0	Guide to the WMO Information System Volume II - WMO Information System 2.0	https://library.wmo.int/idurl/4/69130
RD-1	Provisions for the Transition from the WMO Information System (WIS) 1.0 and Global Telecommunication System to WIS 2.0	https://library.wmo.int/idurl/4/69050
RD-2	WMO Information System 2.0 Strategy	https://library.wmo.int/doc_num.php?explnum_id=4620
RD-3	WIS 2.0: How to define successful transition?	https://wmo-teams.atlassian.net/wiki/spaces/WIS2/pages/301957121/WIS2.0+how+to+define+successful+Transition

3.16.1.2. System overview

System Context

The WIS 2.0 Node will be available to receive data from the upstream data production system, and to serve these data via the Internet to the WIS 2.0 environment.

The WIS 2.0 Node will be available to receive metadata and to serve these metadata via the Internet to the WIS 2.0 environment.

WIS 2.0 Global Brokers will subscribe to the WIS 2.0 Node, and will receive publication messages

when data and metadata become available.

WIS 2.0 Node

1. Selected core and recommended data from upstream systems
2. Notification messages to WIS 2.0 Global Brokers
3. Selected core data to WIS 2.0 Global Caches
4. Subscription messages from WIS 2.0 Global Brokers
5. Remaining core data and recommended data to WIS 2.0 end users

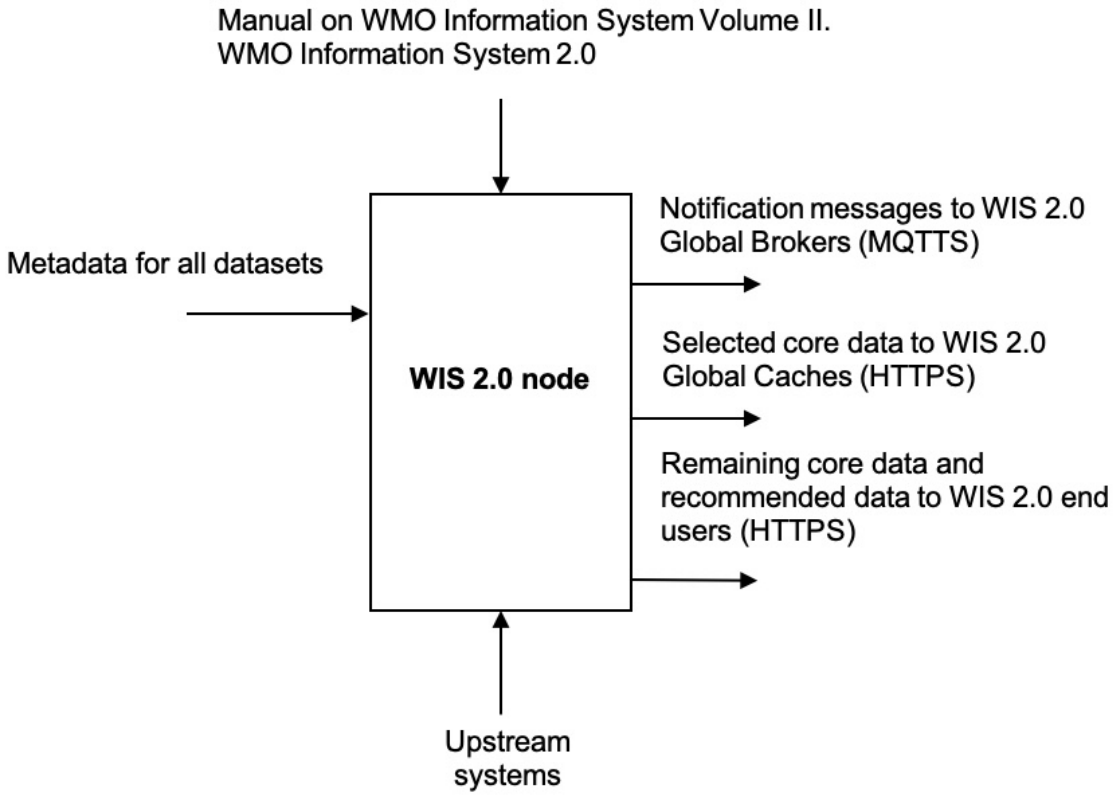


Figure 6. WIS 2.0 Node context diagram

External Interfaces

As identified in the context diagram above, the WIS 2.0 Node will have external interfaces with WIS 2.0 Global Brokers, and WIS 2.0 Global Caches. In each case, the communications will be via the Internet.

The Global Brokers will subscribe to the MQTT broker on the Node. This will be done using the standard MQTT secure port, 8883.

The Global Caches will retrieve core data from the Node by accessing the HTTP server on the Node.

End users will retrieve core data not retained by the Global Caches from the Node by accessing the HTTP server on the Node. End users will also retrieve recommended data from the Node by accessing the HTTP server on the Node, subject to appropriate access control.

The WIS 2.0 Node will be compliant with the Manual on WMO Information System Volume II. WMO Information System 2.0 [AD-0].

3.16.1.3. User requirements

The purpose of the WIS 2.0 Node (hereafter, referred to as the Node) is to be available to receive data from the production system, and to serve these data via the Internet to the WIS 2.0 environment.

In order to reach this target, the following high-level user requirements need to be fulfilled:

USR-0001

The Node shall comply with the specification given in the Manual on WMO Information System Volume II. WMO Information System 2.0, [AD-0].

USR-0101

The WIS 2.0 Node shall include an MQTT broker, using MQTT 5.0 (hereafter, referred to as the Broker).

USR-0150

The Broker included in the Node shall be accessible via MQTT protocol over the Internet.

USR-0201

The Node will allow subscriptions from the WIS 2.0 Global Brokers.

USR-0210

The Node will restrict subscriptions to the Broker by only WIS 2.0 Global Brokers by filtering their incoming IP addresses as made available by WMO Secretariat.

USR-0220

Access to the Broker shall be password controlled.

USR-230

The secure version of MQTT (MQTTS) shall be used. The use of SSL certificates to support this shall be maintained over the lifetime of the system.

USR-240

The Broker shall publish messages using MQTT's Quality of Service (QoS) level 1. This is defined as follows: "*The broker/client will deliver the message at least once, with confirmation required.*".

USR-0301

Upon the arrival of data for distribution via WIS 2.0, the Broker shall publish an MQTT message

announcing the availability of the data.

USR-0320

Notification messages published by the Node shall be formatted in geoJSON, in accordance with the Manual on WMO Information System Volume II. WMO Information System 2.0, [AD-0].

USR-0340

Notification messages published by the Node shall be published using an MQTT topic defined in accordance with the WIS 2.0 topic hierarchy.

USR-0360

The MQTT topic used in notification messages shall not be configured to retain messages.

USR-0401

The Node shall provide access to core data via HTTP over the Internet.

USR-0450

The Node shall provide access to recommended data via HTTP over the Internet, subject to appropriate access control.

USR-0500

The Node shall provide access to core data via HTTPS over the Internet.

USR-0550

The Node shall provide access to recommended data via HTTPS over the Internet, subject to appropriate access control.

USR-0601

The Node shall support the retrieval core data by the WIS 2.0 Global Caches.

Extract from the Manual on WMO Information System Volume II. WMO Information System 2.0 [AD-0]:

FUNCTIONAL REQUIREMENTS OF A WIS NODE

3.6.1 General

3.6.1.1 A WIS Node is the component that enables an NC or DCPC to publish their data and discovery metadata via WIS.

3.6.1.2 See also 3.3 (Functional requirements of an NC) and 3.4 (Functional requirements of a DCPC).

3.6.2 Provide access to data and discovery metadata

3.6.2.1 A WIS Node shall provide access to data in accordance with the WMO Unified Data Policy (Resolution 1 (Cg-Ext-2021)).

3.6.2.2 A WIS Node shall allow one or more Global Caches to access and download core data it publishes for real-time and near real-time exchange. Global Caches provide highly available access to copies of these resources.

3.6.2.3 A WIS Node may restrict access to its core data, relying on Global Caches providing access to data consumers.

3.6.2.4 A WIS Node may provide access to data using a Web-based Application Programming Interface (API).

3.6.2.5 A WIS Node shall provide access to discovery metadata describing the data it makes available and how that data can be accessed. Discovery metadata from a WIS Node is added to the Global Discovery Catalogue to create a consolidated view of data available from all WIS Nodes.

3.6.2.6 A WIS Node shall have the capability to publish notifications via a Message Broker.

3.6.2.7 A WIS Node shall publish notifications via its Message Broker about updates to the data and discovery metadata it provides – including the availability of new data, changes to discovery metadata, and removal of a data set from WIS.

3.6.2.8 A WIS Node shall use a standardized topic structure when publishing notifications. Note: More information on the standardized topic structure is provided in the Guidance on technical specifications of WIS 2.0.

3.6.2.9 A WIS Node shall allow one or more Global Brokers to subscribe to notifications published via its Message Broker. Global Brokers provide highly available distribution of notifications published by a WIS Node.

3.6.2.10 See also 4.3 (WIS-TechSpec-2: Publishing data and discovery metadata).

Note: More information on the function and implementation of a WIS Node is provided in the Guidance on technical specifications of WIS 2.0.

3.6.3 Monitor performance of a WIS Node

3.6.3.1 Each WIS Node shall contribute to monitoring the performance of WIS.

3.6.3.2 See also 4.7 (WIS-TechSpec-6: Managing operations of the WIS).

[1] http://codes.wmo.int/wis/topic-hierarchy/earth-system-discipline/weather/aviation

[2] https://www.go-fair.org/fair-principles/a2-metadata-accessible-even-data-no-longer-available/

[3] https://standards.iso.org/iso/19115/-3/mcc/1.0/codelists.html

Chapter 4. Recipes for Earth system discipline domain experts

4.1. Defining and proposing topic for WMO Earth system disciplines

The WIS2 Topic Hierarchy describes the topic structure and levels to be used when publishing WIS2 notification messages.

The purpose of this document is to provide guidelines to domain experts so that the Topic Hierarchy definition is consistent and useful to address the needs of WIS2 users.

4.1.1. Core levels

Core topics are defined in the first 7 levels and address all Earth system disciplines in a consistent manner.

Core topic examples:

- `cache/a/wis2/ma-marocmeteo/core/data/weather`
- `origin/a/wis2/de-dwd/core/data/ocean`
- `cache/a/wis2/jp-jma/core/data/weather`

The Earth system disciplines defined by WTH are as follows:

- `atmospheric-composition`
- `climate`
- `cryosphere`
- `hydrology`
- `ocean`
- `space-weather`
- `weather`

Topics within each Earth system discipline are then defined by domain experts, reviewed, approved, and published by WMO.

4.1.2. Additional Earth domain specific topics levels

4.1.2.1. What is the use of the Topic Hierarchy ?

The definition of the Topic Hierarchy is heavily linked to the use of Publish-Subscribe (Pub/Sub) protocols, here MQTT[S], in WIS2. Users, when subscribing to the Global Broker, can decide which notification messages they wish to receive.

4.1.2.2. MQTT wildcards

For example, connecting to a Global Broker and subscribing to:

`cache/a/wis2/+/core/data/weather/surface-weather-observation/synop`

users will receive a notification when a new synop is made available from **any** centre-id publishing message on WIS2. The `+` character is a single level wildcard for MQTT subscription.

A user can also choose to subscribe to:

`cache/a/wis2/de-dwd/core/data/weather/#`

In this case, users will receive notification messages from the `de-dwd` centre-id for **all** weather data. The `#` character is a multiple level wildcard for MQTT subscription, and can only be used at the end of a topic subscription.

By using `+` and `#` (the two defined wildcards in the MQTT[S] protocol), users can extend their subscription and receive all messages they need.

The purpose of additional topic hierarchy levels is to allow users to get the messages they need and to be more specific in their request.

4.1.3. Dos and don'ts when defining Earth system discipline topics

Define only a *small number* of additional levels

According the the MQTT[S] protocol specification, the accepted length of a given topic is more than 65kB. This means that the number of levels can be extremely large. However, as explained in https://www.emqx.com/en/blog/advanced-features-of-mqtt-topics:

Try not to use more topic levels "just because I can". For example, `my-home/room1/data` *is a better choice than* `my/home/room1/data`.

By default, some MQTT brokers are configured to accept a maximum of 10 levels. This can be changed in the configuration of the broker, however, this limit shows that a usable, practical topic structure should not be too deep (i.e. with a large number of levels).

For WIS2, and considering the various Earth system disciplines, a limit of **4** sublevels seems appropriate.

Do not use the topic as a metadata record

When defining the topic, experts must focus on the needs of users. The purpose of the WIS2 Topic Hierarchy is **not** to describe as precisely as possible what data users will obtain if they decide to subscribe and then download. Rather, it is *only* to provide a filtering mechanism so that users will not be flooded by WIS2 Notification Messages that may not be useful for them.

In WIS2, all datasets **must** be described using the WMO Core Metadata Profile version 2(WCMP2). Users will be able to discover the data they need by searching the WIS2 catalogue using (via search engines, directly, or from portals and applications). The topic hierarchy information will be part of the metadata record for data which provides real-time notifications of publication.

*Do not allow locally defined sublevels outside the `experimental` topic

Each Earth system discipline provides an `experimental` topic. For example, for `weather`, the first additional levels in the domain topic hierarchy are:

Name	Description
`advisories-warnings`	Advisories and warnings
`aviation`	Aviation
`prediction`	Data sets produced by quantitative algorithms, such as numerical or statistical prediction models, describing the past, present and future meteorological states
`space-based-observations`	Space based observations
`surface-based-observations`	Surface based observations
`experimental`	Experimental topics

As the name suggests, `experimental` allows for defining additional topics for tests and experiments. This is not meant to be used for operational data exchange. It should only be used for testing purposes.

With the exclusion of the `experimental` topic level, the topic Hierarchy must be *fully* defined for each Earth system discipline.

In some situations, it might be tempting for a data producer to use additional topic levels to restrict even more the number of messages received by the users.

This is **forbidden**.

In each Earth system discipline community, all WIS Centres will be able to use the entire topic hierarchy of the domain if they provide data corresponding to each topic. A WIS Centre will not be allowed to add additional sublevels or undefined level within the Topic Hierarchy for its own needs.

Modification of the Topic Hierarchy will be possible by using the WMO fast-track approval process.

Consider users needs and prevent complex wildcard subscriptions

The purpose of the WIS2 Topic Hierarchy is to inform users about the availability of new data. In WIS2, obtaining data will start, in most cases, by configuring one or more subscription to topics, as defined in the associated WCMP2 discovery metadata records, so that users will receive notifications when new data is available.

The Topic Hierarchy should be defined so that users will not need to configure a very large number of different subscriptions to get the data they are interested in.

Each level in the WIS2 Topic Hierarchy should be seen as a "logical" group (as the Earth system disciplines `weather`, `ocean`... or like `synop` for `surface-based-observations`).

Then, and considering that wildcard subscription (using `+` and `#` as described above), are "expensive" to manage for the brokers.

For example, a topic hierarchy resulting in users subscribing to:

```
cache/a/wis2/+/core/data/ocean/+/some/+/thing/+/else/#
```

should be avoided. A subscription to the following topic:

```
cache/a/wis2/+/core/data/ocean/some/thing/else/#
```

is much more effective for both the client and the broker side.

If it is likely that most users will use wildcards for particular topic levels, then, either removing that level altogether, of moving that level a the end of the topic hierarchy is also more efficient for clients and producers.

If most users end up subscribing to:

```
cache/a/wis2/+/core/data/ocean/+/thing/+/#
```

then, the Topic Hierarchy could be reconsidered, so that the above subscription can be replaced by:

```
cache/a/wis2/+/core/data/ocean/thing/#
```

Reordering the levels of topics and potentially reducing the number of sublevels makes the topic hierarchy simpler and more efficient.

Facilitate client side filtering

Notification messages are small pieces of information. MQTT[S] broker and clients are able to handle a very large number of messages. In that sense, receiving, potentially, too many messages is not a problem. However, downloading data, depending on the size of the data might be slower and less efficient. If, for a particular dataset, the geometry information available in the notification message is not sufficient to allow client-side filtering before download, it is suggested to provide additional information in the `properties` object of the notification message so that users can decide *before* downloading if the data in this particular messsage is useful for them.

See [_advertise_client_side_filters_for_data_subscriptions_in_wcmp2_and_wnm] for more information on client side filtering

4.2. WCMP2 Extensions: creating Earth system discipline domain specific profiles of WCMP2

OGC API - Records - Part 1: Core: Requirements Class: Record Core allows for extension and profiling of the metadata content model. In this context, WCMP2 itself is a domain profile, as WMO's discovery metadata standard for all Earth system disciplines.

As part of WIS2, Earth system discipline domain experts may choose to extend and profile WCMP2 to meet their specific needs. Typical examples of where extensions can be useful include, but are not limited to:

- further constraining specific WCMP2 properties (e.g. based on a controlled vocabulary)
- enforcing optional WCMP2 properties to be required
- adding new or additional properties to a WCMP2 record

Profiling WCMP2 can enable domain communities achieve deeper interoperability (or tighter coupling) to meet their needs. For example, an NWP profile of WCMP2 may drive a user-focused NWP portal which leverages domain specific properties of a WCMP2 record for more meaningful search results and assessment for use.

4.2.1. Considerations for creating a WCMP2 Extension

Before creating an extension, domain experts need to give thought on the following aspects:

- gap analysis in WCMP2: can existing WCMP2 properties be used?
- use cases: consider how the additional properties (or constraints on existing) would be used (as a queryable, returnable, etc.)
- sustainability: while developing a WCMP2 extension can be worthwhile metadata modelling exercise, ensure that there is a team in place to help maintain the extension over time. This means having clear ownership of a WCMP2 extension that is subject to ongoing review, and supporting user questions, issues, enhancements and bug fixes

4.2.2. How to create a WCMP2 Extension

WCMP2 extensions can be defined on WMO's GitHub repository for WCMP2 Extensions. Domains can identify and propose extensions as GitHub issues and Pull Requests, which are reviewed by WMO TT-WISMD:

The following elements are required for any WCMP2 extension:

- directory on GitHub repository, containing:
  - examples (directory `examples/`): a directory of example WCMP2 records exemplifying the extension
  - JSON Schema (directory `schema/`): a directory of the JSON Schema definition for the extension, encoded as YAML. The JSON Schema definition should either refer to controlled vocabularies from existing online vocabularies or define inline as `enum` lists
  - README (file `README.md`): the WCMP2 Extension Specification, which includes:
    - owner: clear, unambiguous ownership identification of the Extension
    - prefix: prefix for all defined elements (i.e. `hydro:`, `nwp:`, etc.)
    - conformance: URI for conformance identification and usage in WCMP2 documents (`conformsTo` member)
    - maturity: level of maturity guideline for usage
    - definitions: specification of additional elements and their requirements (required/optional)

Example directory structure

```
|__hydro
|____README.md
|____schema
| |____hydro-schema.yaml
|____examples
| |____example1.json
| |____example2.json
```

An example extension can be found at https://github.com/wmo-im/wcmp2-extensions/tree/main/example.

4.2.3. Differences in WCMP2 records with Extensions

A WCMP2 record with a defined extension per above would provide the following in `conformsTo`:

Example of identifying additional conformance for a WCMP2 Extension

```
"conformsTo": [
    "http://wis.wmo.int/spec/wcmp/2/conf/core",
    "https://wmo-im.github.io/wcmp2-extensions/hydro"  # TODO: clarify how to best
identify
]
```

This will allow a WCMP2 parser to detect additional conformance to a given Extension and validate accordingly (in addition to validating WCMP2 Core).

Annex A: Revision History

Date	Release	Proposed by	Primary clauses modified	Approved by
June 2026	1.1.0	SC-IMT	Addition/update of recipes	SC-IMT
December 2025	1.0.0	SC-IMT	Initial approval for publication	SC-IMT